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Palaeomagnetic Results on Gondwana Dykes from the Damodar Valley Coal-fields and their Bearing on the Sequence of Mesozoic Igneous Activity in India

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Summary

Detailed palaeomagnetic measurements have been carried out on twelve dolerite dykes intruding a belt of Gondwana coalfields exposed in the Damodar valley region and stretching between the Rajmahal traps on the east and the Deccan traps on the west. The problem of age and correlation of the dolerite dykes activity with either the Rajmahal or the Deccan traps activity has been examined in the light of the palaeomagnetic data on these formations. The palaeomagnetic data suggests that the Rajmahal–Sylhet group of traps on one hand and the Deccan traps on the other do not represent independent igneous activities as was hitherto believed and that they are phases of a continuous Mesozoic igneous activity which began on the eastern side with a more or less contemporaneous eruption of lavas giving rise to the Rajmahal, Sylhet and Rajmahendri traps, and which passed through a hypabyssal phase giving rise to the coalfield dykes to the west of Rajmahal traps and then again manifested itself further west in the vast sequence of lava flows constituting the Deccan traps.

Introduction

Palaeomagnetism has proved to be an important tool in the stratigraphic classification and correlation of rock-formations, especially in cases where fossil evidence is lacking. Such a situation is generally met with in the case of igneous activities of uninterrupted nature. Many instances of the use of palaeomagnetic characteristics of rocks in delineating the stratigraphic sequence of different geological formations have been reported in literature (Irving 1964). In this paper, the palaeomagnetic results on a suite of dolerite dykes, intrusive into Gondwana basins forming a part of the Damodar Valley coalfields of India have been reported, and the inferences based on these results regarding the interrelationship of four different Mesozoic volcanic activities in India are discussed.

Geology and palaeomagnetism of the Mesozoic volcanics

The Mesozoic volcanic activity in India is represented by four different outcrops of basaltic lavas (Fig. 1). Of these the Deccan traps have a relatively much bigger extent and their present residual area is about 200 000 square miles. The formation next to Deccan traps in areal extent is the Rajmahal traps. Two comparatively smaller outcrops are the Sylhet traps in Assam and the Rajamahendri traps in Andhra Pradesh. The Deccan traps volcanic activity is supposed to have begun in the east

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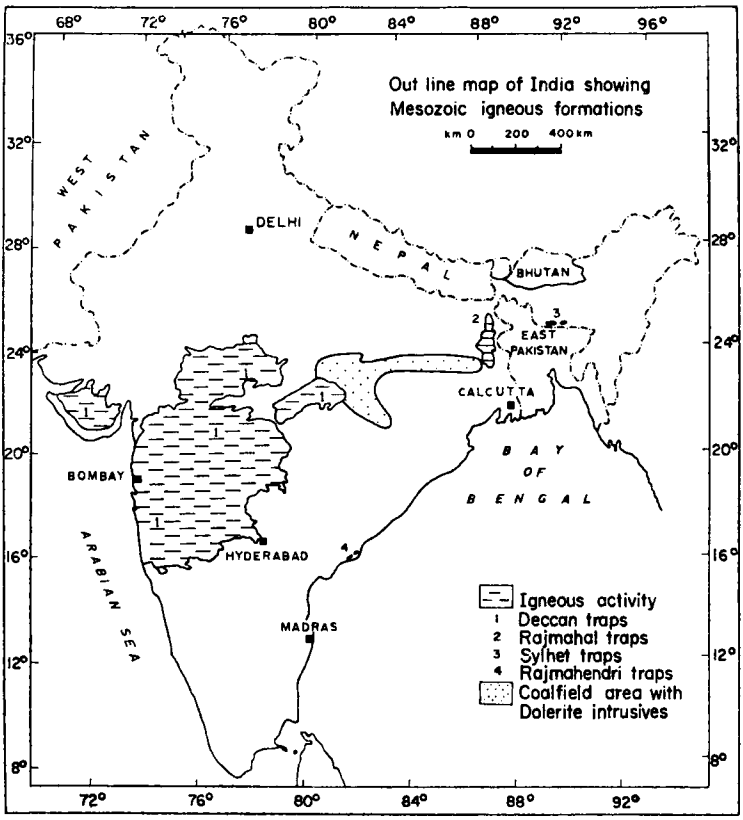


FIG. 1. Map of India showing areal extent of Plateau basalt outcrops. The outlined area between Rajmahal traps (2), and Deccan traps (1) is made of a series of Gondwana period sedimentary basins intruded by Dolerite dykes.

and extended westwards up to the coast where the traps attain a thickness of about 6000 ft (Pascoe 1964). The age span of Deccan traps is taken as from Upper Cretaceous to Lower Eocene (Wadia 1961). The Rajmahal traps have been assigned an Upper Jurassic age on the basis of plant fossil evidence coming from intertrappean beds. However, as will be discussed later, this fossil evidence is not conclusive and the Rajmahals could be of much younger age.

No reliable fossil evidence is available regarding the age of the Sylhet traps and the Rajamahendri traps. However, in Indian geological literature these have been conventionally considered as equivalents of the Rajmahal traps and Deccan traps respectively (Krishnan 1960).

Palaeomagnetism of the Deccan traps, Rajmahal traps, Sylhet traps and Rajamahendri traps has been studied in detail by various workers. The available palaeomagnetic data on these formations is summarized in Table 1. Also given in the table are the palaeolatitude of a central Indian town (Nagpur—21° N, 79° E) and the pole positions calculated from the remanent magnetic directions. The table also gives data on some sedimentary formations from India whose palaeomagnetism has been studied. (Athavale *et al.* (1969a)).

As can be seen from the table, the Rajmahal, Sylhet and Rajamahendri traps are normally magnetized while the Lower Deccan traps are reversely magnetized and the Upper Deccan traps are normally magnetized.

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Table 1

Palaeomagnetic results on some Indian formations

Formation	Age	Mean Remagnetic directions D	I	Polarity	α_{95}	Palaeolatitude of Nagpur	Palaeomagnetic pole Lat.	Long.	Reference
(1) Kamthi sandstones	Upper Permian	306	−50	N	2°	29° S	18° N	53° W	Verma & Bhalla (1968)
(2) Parsora sandstones	Triassic	318	−40	N	2·5°	23° S	30° N	55° W	Bhalla & Verma (1969)
(3) Rajmahendri traps	Upper Jurassis to Lower Cretaceous	307	−47	N	4°	25° S	22° N	48° W	Athavale *et al.* (1969)
(4) Sylhet traps	Upper Jurassic to Lower Cretaceous	332	−59	N	7°	35° 30′ S	16° N	60° W	Athavale *et al.* (1963)
(5) Rajmahal traps	Upper Jurassic to Lower Cretaceous	323	−64	N	4°	44° S	13° N	69° W	Radhakrishnamurty (1963)
(6) Satyavedu sandstones	Lower Cretaceous	321	−58	N	4°	33° S	26° N	67° W	Mital *et al.* (1970)
(7) Tirupati sandstones	Lower Cretaceous	153 323	+56 −59	R N	4°	34° S	28° N	73° W	Verma & Pullaiah (1967)
(8) Lower Deccan traps	Middle to Upper Cretaceous	154	+53	R	10°	30° 30′ S	30° N	79° W	McElhinny (1968)
(9) Upper Deccan traps	Upper Cretaceous	345	−44	N	10°	25° 30′ S	37° N	78° W	McElhinny (1968)
(10) Pavagadh basic rocks (Lower)	Eocene	351	−16	N	8°	9° S	58° N	91° W	Deutsch *et al.* (1959)
(11) Pavagadh acid rocks (Upper)	Eocene	351	+17	N	7°	6° 30′ N	75° N	89° W	Deutsch *et al.* (1959)

Geology of the Son-Damodar Valley region

The distance between the north-eastern margin of the Deccan traps and the Rajmahal traps (Fig. 1) is about 300 miles. This intervening stretch consists of a belt of troughs or basins in which rocks belonging to the Gondwana system have been deposited. The Gondwana system in the stratigraphy of India represents a characteristic group of land deposits which range in age from Upper Carboniferous through the greater part of Mesozoic era up to Upper Jurassic.

The exposures of Gondwana rocks are mainly developed along two sides of a great triangular area (Krishnan 1960), the northern side of which is roughly represented by the Damodar, Son and Narmada valleys trending E–W. In Fig. 1, the area intervening between the Deccan traps and Rajmahal traps outlines the isolated Gondwana exposures belonging to this northern group and exposed in the Damodar and Son Valleys. The western side represented by the Narmada river valley is not shown in Fig. 1.

The other prominent branches of the triangle are developed along the NW to SE trending Godavari valley and along the Mahanadi valley in the east.

Hypabyssal igneous activity giving rise to dolerite dykes and sills is rare in the Mahanadi and Godavari valley exposures (Krishnan 1960) while the Son–Damodar belt of Gondwana coalfields lying between the Deccan traps on the west and Rajmahal traps on the east is infested with numerous dykes and sills. Fig. 2 shows a detailed

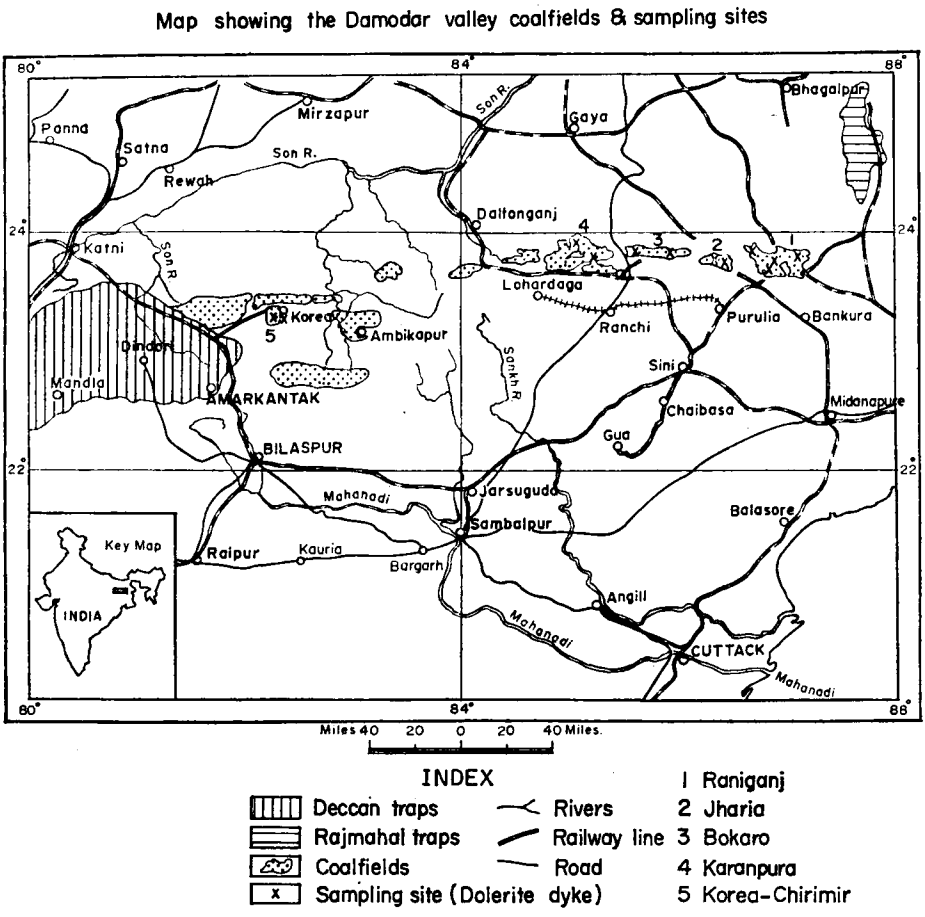


FIG. 2. Map showing the Damodar valley coalfields and sites where dolerite intrusives were sampled for palaeomagnetic studies.

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map of this area. The age of this igneous activity has been a controversial topic in Indian stratigraphy. It has been customary to consider these intrusives as related to the Rajmahal traps activity. However on the score of distance there is an equal probability that these could be fully or partially representative of the Deccan trap volcanism (Fox 1930). The Deccan traps and Rajmahal traps look very similar in hand specimens. Chemical analysis and testing of physical properties have also not brought out any parameters distinguishing these two types of trap-rocks (Pascoe 1964).

Since detailed palaeomagnetic studies have been conducted on both Rajmahal and Deccan traps and these have brought out certain characteristic features which distinguish these two suites of igneous rocks, it was felt that palaeomagnetic studies on the dolerite dykes in the Damodar valley coalfields may be able to answer the question as to whether the dykes represent the Rajmahal or the Deccan traps phase of igneous activity. No radiometric dates were available for the dolerite dykes and the Rajmahal traps at the time when this study was undertaken.

Palaeomagnetism of Gondwana dykes

(a) *Collection of samples*

Oriented blocks of samples were collected from twelve dykes distributed over the region of the Son–Damodar coalfields. On an average, seven block samples were collected from each site. The distribution of sites is given below and is also shown in Fig. 2.

(1) *Raniganj coalfield.* Four dolerite dykes mentioned in Gee's (1932) memoir on this coalfield, namely the Salama dyke, Charanpur dyke, Sitarampur dyke and Barakar river dyke; Sites Nos 14, 15, 16 and 20 respectively. The site numbers are not in serial order as some other sedimentary formations were sampled in the same field-trip.

(2) *Jharia coalfield.* Two dolerite dykes exposed in railway cuttings near Kharkhari and Jamuniatand railway stations and referred to in Fox (1930); Site Nos 9 and 10.

(3) *Bokaro coalfield.* Two dolerite dykes, one exposed at Tapin approachable from Charhi on the Hazaribagh–Ramgadh main road; sampled at two points; and the other exposed near Tilaiya, 7 km from Chainpur. Site Nos. 25A and B and 30.

(4) *Karanapura coalfield.* Two dykes, one exposed in Damodar river near NCDC Sayal ' D ' quarry and the other near Patratu Railway Station, Site Nos 27 and 28.

(5) *Korea-Chirimiri coalfield.* (i) Dyke exposed in road cutting after first Nullah on road from Nagpur to Chirimiri Colliery; Site No. 36. (ii) Dyke exposed in road cutting, three miles from Manendargarh on Ambikapur–Manendargarh road. Site No. 37.

(b) *Measurement of NRM directions and stability tests*

Three to four cylindrical specimens were cored from each of the block samples and their directions of natural remanent magnetism were measured on an astatic magnetometer.

Initially, a few pilot specimens were taken from each site for demagnetization in alternating field at successively increasing peak-field strengths of 25, 50, 75, 100, 150 and 200 Oersteds. By about 200 Oersteds, most of the specimens became very weak in intensity. Based on the change in the remanent vector and the decay in intensity of magnetization of the pilot specimens, a peak-field strength value value of 25, 50, 75 or 100 Oersteds was fixed for the demagnetization of remaining

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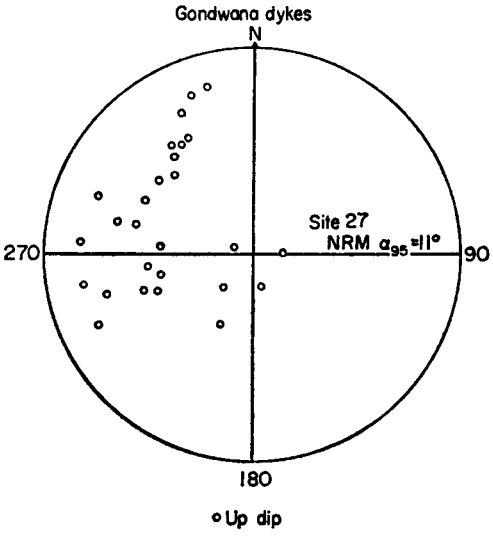


Fig. 3

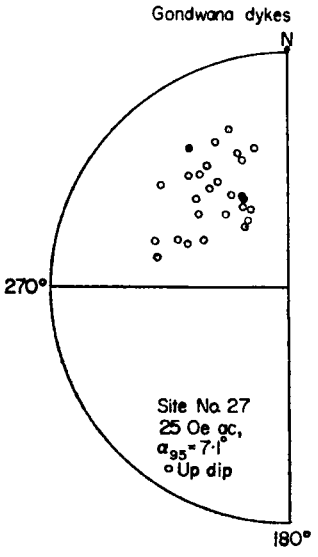


Fig. 4

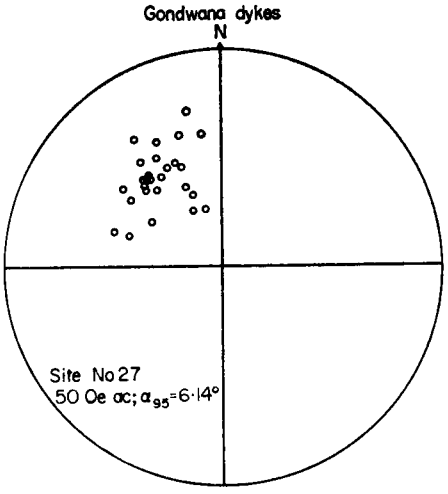


Fig. 5

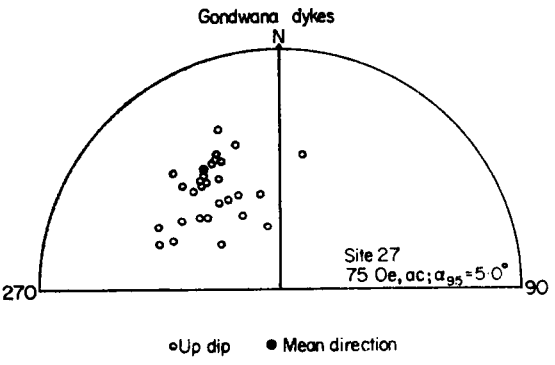


Fig. 6

FIGS 3–6. Stereographic projections showing the remanent magnetization directions for specimens from Site No. 27. Fig. 3: NRM; Figs 4, 5 and 6: on alternating field demagnetization in peak-fields of 25, 50 and 75 Oersteds, respectively.

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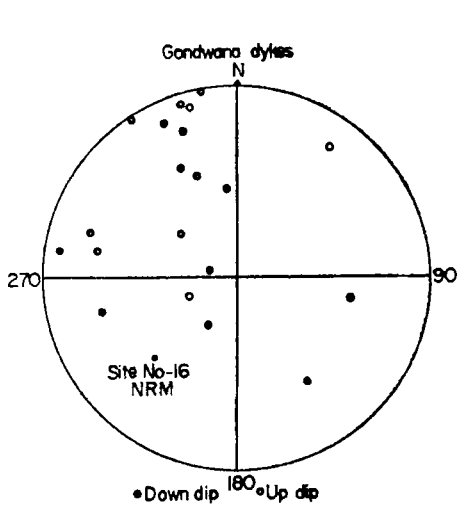


Fig. 7

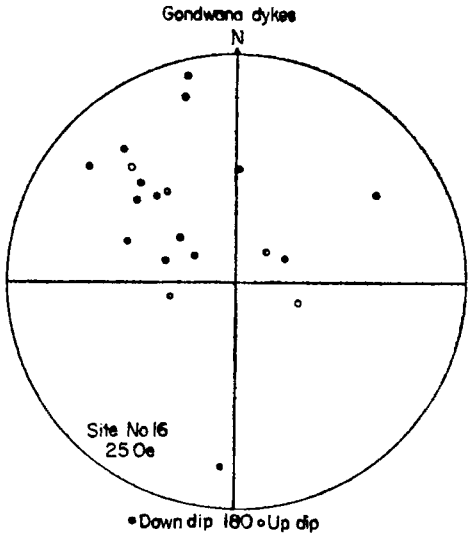


Fig. 8

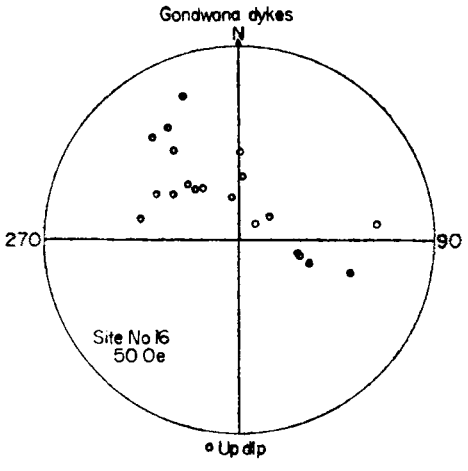


Fig. 9

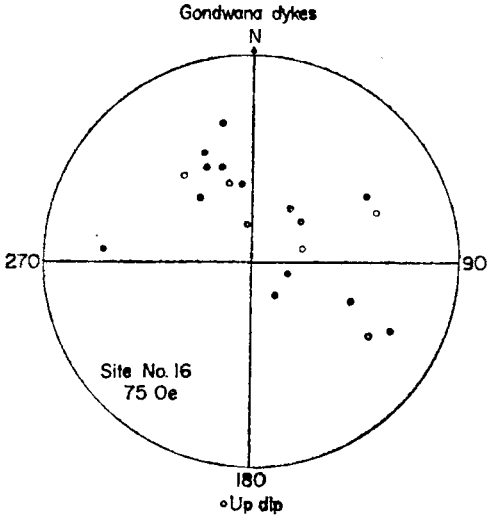


Fig. 10

FIGS 7–10. Stereographic projections showing the remanent magnetization directions for specimens from Site No. 16. Fig. 7: NRM, directions; Figs 8, 9 and 10 on alternating field demagnetization in peak-fields of 25, 50 and 75 Oersteds respectively.

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specimens from each site. Figs 3–6 show the removal of secondary components and reduction in the scatter in the case of one site (No. 27) on demagnetization in successively increasing peak-field strengths. Fisher's (1953) analysis was used for calculating α_{95}, the coefficient of scatter for the NRM directions, as well as for directions obtained after a.c. cleaning in successive peak-fields, of 25, 50 and 75 Oersteds. The α_{95} values obtained at each level are also given in Figs 3–6.

Excepting the case of Site No. 16, the scatter in the NRM directions was considerably reduced in all other sites as a consequence of A.C. cleaning. In the case of Site No. 16, alternating field demagnetization at successively increasing fields did not reduce the scatter in NRM directions as shown in Figs 7–10. A few more specimens were then cored from the block samples of this site and these were subjected to thermal demagnetization in an apparatus (Athavale 1969b) capable of annuling the Earth's magnetic field up to 0·15 per cent of its value over a 4 in. radius spherical volume at the centre of the Helmholtz coil system where the specimens are cooled to room temperature. The stable remanent directions for specimens from Site 16 on thermal demagnetization at 250 °C are shown in Fig. 11.

(c) *Palaeomagnetic Results*

Table 2 gives the final results on the mean stable remanent directions for the twelve sites of dolerite dykes. Also given in the table are the statistical parameters (Fisher 1953) such as precision (K) and circle of confidence at 95 per cent probability level. The palaeolatitude of a central Indian town-Nagpur (21° N, 79° E)—calculated from the palaeomagnetic data in the case of each dyke is also given in the table. It leads to the pertinent conclusion that at the time of the eruption of these dykes, India was located in the Southern Hemisphere. This data thus confirms the observation of earlier workers about continental drift of India, based upon palaeomagnetic studies of igneous and sedimentary rocks. (Athavale *et al.* 1969a).

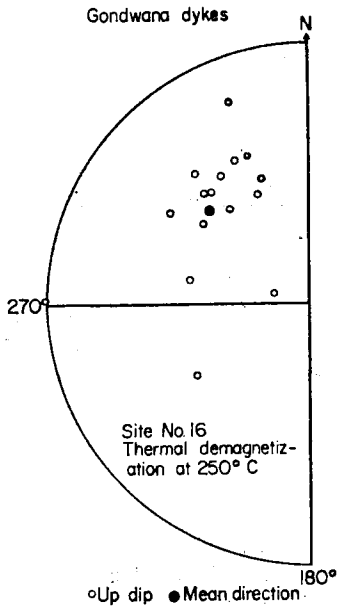


FIG. 11. Remanent magnetization directions for specimens from Site No. 16 on thermal demagnetization at 250 °C.

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Table 2

Palaeomagnetic data on Gondwana dykes

Site No.	Geophysical Co-ordinates Lat; N	Long; E	No. of samples	Mean stable remanent directions D	I	Precision parameter K	α_{95}	Palaeo latitude of Nagpur	Pole position Lat.	Long.	Oval of confidence for pole δp	δm	Coalfield
9	23° 46′	86° 14′	3	350	−28	40	12·7°	15° S	50° N	76° W	7·6°	14°	Jharia
10	23° 36′	86° 14	8	338	−40	41	8°	22° S	39° N	67° W	5·6°	9·3°	Jharia
14	23° 38′	87° 0′	4	326	−58	49	6·7°	37° S	20° N	66° W	7·4°	6·1°	Raniganj
15	23° 38′	86° 58′	6	11	−47	167	4·4°	32° S	37° N	105° W	3·7°	5·7°	Raniganj
16	23° 38′	86° 58′	7	312	−47	67	6·4°	28° S	21° N	48° W	5·4°	8·3°	Raniganj
20	23° 58′	86° 58′	9	329	−56	56	6·2°	43° S	23° N	66° W	6·2°	9°	Raniganj
25	23° 49′ N	85° 54′ E	9	353	−35	34	9°	21° S	46° N	85° W	5·9°	10°	Bokaro
27	23° 36′ N	85° 22′	13	327	−40	60	5°	25° S	33° N	64° W	3·7°	6°	Karanpura
28	23° 36′ N	85° 22′	4	350	−43	38	3·6°	33° S	34° N	86° W	2·8°	4·1°	Karanpura
30	23° 49′ N	85° 54′ E	7	340	−37	27	10°	25° S	41 °N	69° W	10·6°	15°	Bokaro
36	23° 15′ N	82° 15′ E	13	317	−48	77	4°	29° S	23° 30′ N	58° W	3·4°	5·2°	Korea chirimiri
37	23° 15′ N	82° 15′ E	5	312°	−46	114	3°	26° S	22° N	53° W	4·7°	10°	Korea chirimiri
Mean	23° 45′	85°	98	336	−45°	37	6·7°	24° S	33° 30′ N	61° W	5°	5°	

Assuming that the land mass has been stationary and the Magnetic Pole has wandered in the past, the virtual pole positions which could give rise to the observed directions in the dykes have been calculated and these are also given in the table. Assuming that the eruption of all the dykes has been a more or less contemporaneous event, a mean direction of magnetization and mean pole position has been calculated for them using Fisher's (1953) analysis. Figure 12 is a stereographic projection in which the pole position for each individual dyke has been plotted; the mean pole position is also indicated in the diagram.

The virtual geomagnetic pole positions for palaeomagnetic data in Table 1 are plotted in Fig. 13 along with the mean Palaeomagnetic pole for the Gondwana dykes. The Kamthi sandstones (Verma & Bhalla 1968) and the Parsora sandstones (Bhalla & Verma 1969) are the oldest Mesozoic formations studied in India. However, the remanent magnetism in both of these formations is of chemical origin and hence their magnetic age could be much younger. The Sylhet traps are believed to be contemporaneous with the Rajmahal traps and this is borne out by the similarity in their palaeomagnetic poles. The age of the Rajamahendri traps is a controversial issue and fossil evidence is not unambiguous (Pascoe 1964). On the basis of palaeomagnetic evidence, Radhakrishnamurthy (1963) has classified them as contemporaneous with the Rajmahals. Bhimshankaram (1965) reports almost similar palaeomagnetic directions, but treats them as of upper Deccan trap age. An examination of Fig. 13 suggests that their pole position is closer to pole positions for Sylhet and Rajmahal traps than to that for the Upper Deccan traps.

Discussion on the age of the Gondwana dykes igneous activity

A polar wandering curve is drawn through the various pole positions in Fig. 13, on the basis of the age sequence of the various formations.

The Deccan traps is the most widely studied formation and the pole positions for upper and lower Deccan traps represent the mean of 92 sites of lava flows and dykes in the Deccan trap country (McElhinny 1968).

The Rahmahendri–Sylhet–Rajmahal traps igneous activities also represent a well-defined group in the SE corner of the diagram. The mean pole positions of the Gondwana dykes is quite different from both these groups.

Several interesting possibilities arise regarding the age of the dykes which are normally magnetized. These are as follows:

(1) They belong to the Rajmahal traps phase of igneous activity.

(2) They belong to the late phase of the igneous activity of the Deccan traps, during which the normally magnetized upper Deccan traps were erupted.

(3) Some of the dolerite dykes belong to the Rajmahal phase of igneous activity while others belong to the Deccan traps phase.

(4) The dykes represent an igneous activity different from that of the Rajmahal traps or the Deccan traps and probably having an intermediate age.

(5) The Rajmahal traps and Deccan traps are manifestations of one single continuous activity of which the intrusion of Gondwana dykes was a part and parcel. The difference in the mean pole positions of the three is perhaps due to rapid rate of continental drift or polar shift during this volcanic period.

These possibilities are now considered one by one.

The palaeomagnetic mean pole position of the Gondwana dykes is different from that of the Rajmahal traps or the Deccan traps. It falls at an intermediate position on the polar wandering curve in Fig. 13. Therefore, possibilities (1) and (2) suggesting

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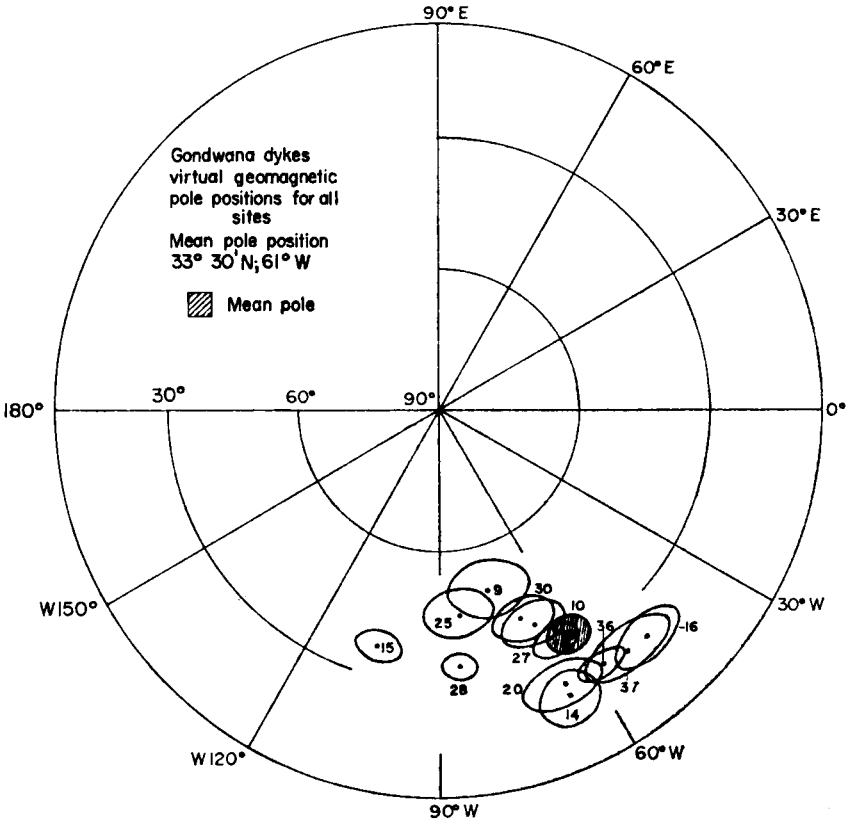


FIG. 12. Virtual geomagnetic pole positions for twelve Gondwana dolerites. The mean pole position is also shown in the diagram.

1 Kamthi sandstones
2 Parsora sandstones
3 Rajmahal traps
4 Sylhet traps
5 Rajmahendri traps
6 Gondwana dykes
7 Satyavedu sandstones
8 Tirupti sandstones
9 Lower Deccan traps
10 Upper Deccan traps
11 Pavagadh basic rocks (lower)
12 Pavagadh acidic rocks (upper)

Polar wandering curve for India

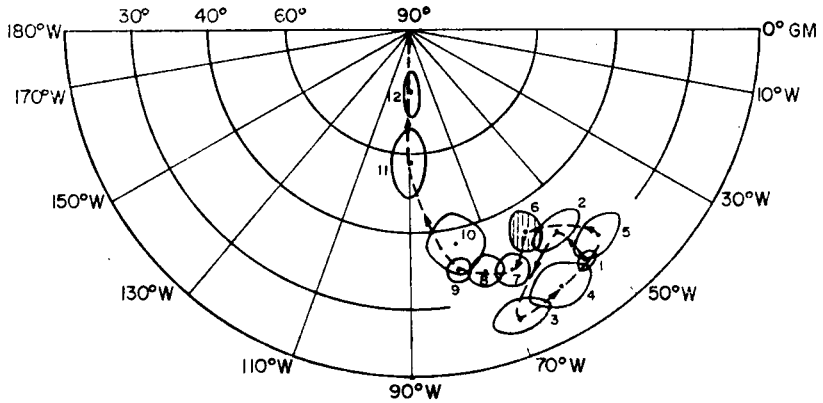


FIG. 13. Virtual geomagnetic pole positions for various sedimentary and igneous formation in Table 2 and the mean pole positions for Gondwana dykes.

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that all the dykes belong exclusively to either the Rajmahal or the Deccan traps igneous activity are ruled out on the basis of palaeomagnetic evidence.

The pole position of three dykes from Raniganj coalfield (Site Nos 14, 16 and 20; Fig. 12 and Table 2), which is geographically very near to the Rajmahal traps, are also close to the mean pole for Rajmahal traps. Therefore, these dykes could be considered as related to the Rajmahal traps activity. However, it is noted that the pole positions of dykes from the Korea–Chirimiri coalfield (Site Nos 36 and 37), which is located very close to the Deccan traps (see Fig. 2), also fall very much nearer to those of Rajmahal and Sylhet traps; and at the same time pole positions of some dykes for the other coalfields nearer to Rajmahal traps (see Fig. 2) like the Bokaro (Site Nos 25 & 30) and the Jharia (Site Nos 9, 10) coalfields fall nearer to those for the Deccan traps. Another point of some statistical significance is that although the pole positions of the dykes show a good deal of internal scatter (Fig. 12) not a single dyke has a pole position which falls within the ovals of confidence of either the Deccan or the Rajmahal traps, if Fig. 12 is superposed on Fig. 13. Thus, although the third possibility cannot be entirely ruled out, there is not enough justification on the basis of palaeomagnetic evidence or geographical proximity to presume that some dykes are of the Rajmahal period and the others belong to the Deccan trap period.

It is rather difficult to decide between possibilities (4) and (5). However assuming that possibility (4) is true, i.e. the dykes are representatives of an independent activity, an upper limit can be given to the age of the dyke system on the basis of Palaeomagnetic data for other formations listed in Table 1. The Lower Cretaceous Tirupati sandstones in Table 1 can be considered to be approximately contemporaneous to the Deccan traps. As can be seen from Fig. 13 the pole position of Tirupati sandstones also falls very close to that of the Lower Deccan traps. Now, in the Deccan trap sequence, the normal traps overlie the reversely magnetized traps, while in the Tirupaties, the reversely magnetized rocks overlie the normal ones (Verma & Pullaiah 1967). We can therefore, conclude that the Tirupati sandstones are slightly older than the Deccan traps and that the two formations represent two successive events of reversal of the Earth's magnetic field and that the lower Deccan traps and the upper Tirupaties could represent the same period of reversal of the Earth's magnetic field.

On the basis of stratigraphic evidence, the Satyavedu sandstones in Table 1 are correlated with the Tirupati sandstones (Krishnan 1960). The Satyavedus are normally magnetized and their pole position falls between those for Tirupati sandstones and the Gondwana dykes (Fig. 13). Since both Satyavedu sandstones and Gondwana dykes are normally magnetized we can consider these as equivalent of the lower Tirupatis or slightly older. Geological evidence (Verma & Pullaiah 1967) suggests that the Tirupaties are of Lower to Middle Cretaceous age, which can be treated as the upper limit of the age of the independent igneous activity giving rise to the Gondwana suite of dykes.

On the other hand, the spread of the individual pole-positions of the dykes, right from the pole position of Rajmahal traps up to that of the upper Deccan traps suggests a continuous period of igneous activity, which is the fifth possibility. Possibility (4) discussed above is valid only on the assumption that the calculation of a mean pole for all the dykes is justified on a physical basis. If however, the dykes have been erupted over a period during which relatively rapid continental drift or polar wandering was taking place, then the spread in their pole positions is not due to secular variation alone and the calculation of a representative mean is not justified. Compared to the relatively short time span between the Rajmahal and the Deccan traps (Pascoe 1964), the difference in their respective mean pole positions, as in Fig. 13, is considerable, indicating that this period of intense volcanic activity was also marked by a rapid rate of continental drift and/or polar wandering.

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If such is the case, then the hypothesis outlined in (4) may not hold good (since the interpretation suggested therein is based on the location of the mean pole for dykes on the polar wandering curve). The present palaeomagnetic evidence thus appears to be more in favour of the fifth possibility.

Opinion of geologists also seems to support this possibility. Pascoe (1964, pp. 1399–1400) says: ' Further investigations may eventually prove that the Rajmahal traps represent but the initial phase of the Deccan disturbance. Between the Rajmahal hills and the Deccan traps, dykes of dolerite, petrologically identical with sills and dykes of Deccan trap are frequent in the Karanpura, Jharia and Raniganj and other coalfields. The fact that the rocks of the two denominations (Rajmahal traps and Deccan traps) are lithologically indistinguishable from each other has always been a subject of remark and authorities have refrained from classifying them as one, only by reason of the wide difference between the supposed Jurassic age of Rajmahals and Late Cretaceous age of the Deccan traps. Recent researches however, tend to lessen the gap. On the evidence of Ammonites, the Raghavapuram and other plant beds (Raghavapuram shales are immediately overlain by Tirupati sandstones—Author) belong to the Lower Cretaceous (upper Neocemian). These plant beds have always been regarded as not much younger, if at all, than the Rajmahal series with which they have many species in common. The Rajmahals therefore, seem to demand a position either at the base of the Cretaceous, or at the top of Jurassic.'

Pascoe (1964) further gives the example of Panjal traps of India and the volcanic activity of Northern Europe represented by the basalt flows of Antrim, Skye and other regions and points out the evidence in them of the tendency to change their locality, beginning in one area and finishing in another. On the basis of these observations, he comments: ' It would be no strain on our credulity therefore, to suppose that the Deccan disturbance began in the Rajmahal hills, spread westwards through the country where it is impossible to distinguish between Rajmahal dykes and Deccan trap dykes, and ended either in Bombay or in some region further west now covered by the Arabian sea.'

The palaeomagnetic data on Gondwana dykes strongly supports the opinions of Pascoe (1964) quoted above.

Conclusions

On the basis of the palaeomagnetic study of Gondwana dykes, the following conclusions may be arrived at:

(1) The Rajmahal and Sylhet traps on the one hand and the Deccan and Rajmahendri traps on the other hand are not manifestations of two independent igneous activities, as was hitherto believed.

(2) The Mesozoic (Jurasso–Cretaceous) volcanic activity in India began on the eastern side with more or less contemporaneous eruption of lavas giving rise to the Rajmahal, Sylhet and Rajamahendri traps, passed through hypabyssal phase of intrusions of dolerite dykes in the coalfields west of Rajmahal traps and then again manifested itself further west in the vast sequence of lava flows of Deccan traps.

(3) During the period of this continuous igneous activity, the Earth's magnetic field had reversed twice in its direction, first giving rise to normally magnetized Rajmahal group of traps and Gondwana dykes, then giving rise to reversely magnetized lower Deccan traps and the coeval Tirupati sandstones and again reversing to give rise to the normally magnetized upper Deccan traps.

(4) The data on the palaeomagnetic directions of the dykes agrees with the earlier palaeomagnetic data from India and can be interpreted in terms of the location of India in the Southern Hemisphere, during the period of intrusion of these dykes. It thus lends a strong support to the hypothesis of continental drift.

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